

# DESD INTERVIEW QUESTIONS

## Toshiba

- 1) Introduce yourself in brief.
- 2) Are you good at C and DSA?
- 3) How will you set nth bit of a variable?
- 4) Find mid of linked list(code)
- 5) Do you know tree data structure?
- 6) Find the left most leaf of tree(code)
- 7) What is structure padding?
- 8) Why we use structure padding?
- 9) Find the size of structure  

```
Struct {  
    Char c  
    Double d  
    Int i  
};
```
- 10) How will you allocate memory dynamically in c?  

```
Void *ptr1  
Int *ptr2`
```
- 11) What is size of ptr
- 12) If same then why we use int and why same ?
- 13) What is process and thread?
- 14) How memory will be given to process and thread ?
- 15) What are memory sections of c?
- 16) How will you find element in binary search algorithm?
- 17) Write the code to clear nth bit
- 18) Write the code to set n bits

## Toshiba Technical main

- 19) What are different different microprocessor and...

## Knorr Bremse

- 1) Write syntax for function pointers
- 2) Also concept of array of pointers
- 3) How C is different from embedded C
- 4) Why it called as embedded C
- 5) Write the function to check the variable is Bool(Boolean) or not
- 6) What is the size of this struct  

```
struct {  
    Char a:2 ;  
    unsigned char b:3;  
    char c:1;  
    unsigned char d:5;  
}
```

- 7) How will you access members from pointer of structure (code)
- 8) Write syntax for pass by reference and pass by value
- 9) If my machine is little endian and I wrote code on it and compile it, will it give same output on big endian machine?
- 10) Do I need to change the code or I need to configure machine?
- 11) And more related to C snippet's questions

### **KNOR - -BREMSE**

1. C Compilation Flow
2. Write code for string and reverse without string.h
3. Structure Pointers
4. Array pointers
5. Bitwise operators code
6. Difference between function and micro.
7. Qualifiers
8. C banking based application concepts.
9. Difference between structure and union.
10. Const and pre-processor same or different.
11. All Protocols
12. Regarding my project in depth (for 30min)
13. PWM Concepts
14. If two structures create its objects and access them and how to know which pointer is which structure.
15. Give real life example of union.
16. Typedef and typecasting

### **KNOR - BREMSE**

- 1) What is data structure?
- 2) Explain your project? (block diagram, interfacing, why you have selected these components only)
- 3) What is difference btw union and structure?
- 4) take a structure calculate its size if it contains structure pointer itself?
- 5) take two structure, now on s2 take structure pointer of s1, with the help of instance of s2 how will you assign value to s1 data members using pointer declared in s1?
- 6) Refer to ques 4, what will size in case of union?
- 7) question related to structure padding?
- 8) write a code to find largest num/smallest num in array?
- 9) you have char arr[]="hello" and char \* str ="hello" which will u choose to make "o" as "e"?
- 10) how many test cases required to find whether number is in range or not?
- 11) const int \*ptr and int \*const ptr what will they represent?
- 12) write a macro to calculate array size?
- 13) explain pwm constant?

### **KNORR - BREMSE**

1. Macro to set nth bit
2. Clear the nth bit
3. toggle the nth bit
4. write a function which takes an endianness (big or little) and return the opposite endianness

5. write a function which will return the answer by changing the lsb bit to msb and msb to lsb
6. which will you choose SPI or i2c to choose for any project
7. All protocols
8. Can we call ISR function in any main.c or .c file
9. Can we give arguments to ISR function
10. How can we access a function written in one file into another file
11. Can we access a .c file into another .c file ? how ?
12. suppose we assigned int \*ptr ; and next line ptr = 3; will it give error or execute successfully
13. what is segmentation fault?
14. What are the storage classes in c ??
15. Function pointer and where is it used ?
16. Interrupt handling and how does it know how to execute ISR for which interrupt n what if we have two interrupt?
17. if we declare a variable in c file how to access it in another file
18. Scheduling algorithm in RTOS.
19. what will happen if interrupt takes longer time than our process n what will happen to waiting process in the queue.
20. stack overflow n why it happens
21. can we use static extern before function return type
22. there is static variable define in c file how will u change that static variable from another file
23. explain the c file compilation flow
24. Difference between macro and const.
25. write a code for reverse the number
26. do the addition of following hexadecimal numbers: 0x91, 0x21
27. can we call ISR function into main.c or any other .c file
28. write a program which take bit and number. now clear the particular bit of that number.

## **ACL Digital**

1. Tell me about yourself
2. About project.
3. Cdac journey
4. Soft skills.

## **Aeron Questions**

Round 1:

1. If a main function containing only printf is called recursively, will it overflow the stack?
2. can we store a double in float? what will happen if we do?
3. what is frequency of Wi-Fi? which frequency has a greater range?
4. how to calculate value of resistors for voltage divider circuit?
5. how to interface LED to microcontroller? how to control its intensity?
6. how to interface relay to microcontroller?
7. what is the diode used in relay called? why is it used?
8. what are pins in ethernet cable?

9. if we transmit 8 bytes over UART, how many total bytes are transmitted?
10. Specifications of supply at home?
11. Can single phase be converted into 3 phase?
12. What is frequency of DC?

Round 2:

1. Tell me about yourself?
2. explain your b.tech project in brief.
3. Explain your CDAC project, why have you used CAN bus in your project?
4. Pins of CAN bus, specifications, draw its frame format, can we directly add a node in between?
5. Explain i2c, is it faster than spi?
6. what is UART/RS232, it's voltage levels?
7. Explain TCP/IP, UDP?
8. Explain work flow of message between client and server socket?
9. Write code to toggle single bit in a number.
10. Write code to multiply a number without using arithmetic operators.

## Thales India Interview Questions

Round 1:

1. What is Scheduling?
2. What is Multi-tasking?
3. What are different types of scheduling algorithms?
4. What is priority scheduling?
5. What is the term called when a low priority process does not get executed? (starvation)
6. How can it be avoided?
7. What is preemption?
8. What is synchronisation?
9. Types of synchronisation?
10. How does semaphores work?
11. Difference between semaphores and mutex?
12. What are device drivers?
13. Name some drivers you have worked upon.
14. How is usb device driver created? Explain its working?
15. What is mounting and unmounting?
16. What is OOPs?
17. What is difference between c & cpp?
18. What are different storage classes in C? explain each class.
19. When is register storage class used?
20. What is volatile keyword? when is it used?
21. Working of volatile keyword.
22. What are macros? why or when are they used?
23. Why are macros used if we already have constant variables?
24. Write a code to check if given string is a palindrome.
25. Write a code to reverse a string without using another string.
26. Write a code to extract number at thousands place from an integer. (ip:12345, op:2)
27. What are bitwise operators?
28. Difference between bitwise AND operation & logical AND operation?
29. Write an assembly code to add two numbers and subtract two numbers.

#### Round 2:

1. What projects did you work on?
2. What programming languages did you use in your projects?
3. Do you know cpp/oops programming?
4. Do you know ADA programming?
5. How quickly can you scale yourself to learn something new?
6. Do you prefer working in a team or individually? which is better?
7. Why do you want to join Thales?
8. What is FMS?
9. What will you do if you need information on something and your manager is not available?
10. Where do you see yourself in the next 10 years?
11. Do you think staying at one company is enough to reach your goal or do you need to switch between different companies?

#### Round 3 (HR):

1. Tell me about yourself? (Personal info)
2. Are you willing to relocate?
3. How quickly are you available for work?
4. Can we expect you to stay at Thales for next 8-10 years?
5. How was your experience with Thales interviewers in last rounds?
6. Are you satisfied with the mentioned salary?

### **Thales India interview questions**

1. Give your introduction.
2. Tell about memory allocations in c.
3. Tell me difference about dynamic and static memory allocation.
4. What is auto variable.
5. What is storage classes and their types
6. What is linked list and double linked list.
7. What is polymorphism
8. Tell me conditions where we use auto and static variable.
9. Write a program to check given number is palindrome or not
10. Write a program to print most repeated element in string.
11. What is RTOS.
12. Write a program to allocate dynamic memory to 1D array and 2D array.
13. Sorting algorithm in DSA explain them.

#### **HR question**

1. Explain how you are a team player.
2. I give you some task which will be complete in next 5 days but you complete it on 3 days then what you will do in rest 2 days.

### **EMBEDOS TECHNICAL INTERVIEW QUESTIONS**

#### **C programming**

1. C compilation flow?
2. In C compilation flow where functions get resolved which are declared in main file?
3. Dangling pointer?
4. Memory Leakage?
5. How to check or detect memory leakage without using any tool?

## **Device Driver**

6. How to load any device driver on each boot automatically?
7. What if we don't want any particular device driver to get loaded at each boot?

## **RTOS**

8. What is priority inversion?
9. How to solve this problem?
10. Which RTOS you worked on?
11. Types of RTOS?

## **OS**

12. What is OS?
13. functionality of OS?

## **Basic Electronics**

14. Red Blue Red Gold colour code resistor value?  
-What is tolerance for given resistor?  
-What is meant by tolerance?
15. Voltage divider ckt, formula?
16. What if we add multiple resistors at load of Voltage divider ckt?
17. Difference between zener diode and simple diode?
18. What if I am getting 3.3 v at output of rectifier circuit can we get regulated signal with the help of zener diode?  
What if I want 5v regulated output? will zener diode will help me to get that?
19. RAM Vs ROM?

## **MCPI**

20. How to set, reset, toggle particular bit of microcontroller?
21. Address space for 32bit Microcontroller and 34-bit microcontroller?

## **GS Lab Technical Interview Questions**

### **MCPI**

1. Which microcontroller you worked on?
2. Have you read data sheet of it? What kind of information is there in datasheet?
3. STM32 registers?
4. Check 13th bit of given 32-bit data and whatever status of that bit (1 or 0) assign it to flag?

### **C programming**

- In which programming language you are comfortable?
5. In given int array of size 150 there are elements which are from 1-9, print occurrence of each number?

### **OS**

6. On which Linux you worked on?
7. Tell Linux Commands you know?
8. What is system calls?
9. Tell system calls you know?
10. Linux command to check memory space available/free?
11. What is difference between RAM and Storage?
13. Linux command to check memory usage?
14. Linux command to check how many CPUs are there on system?

## **HR Questions**

### **self-introduction**

1. code on pyramid
2. code on 4\*4 matrix to find addition of diagonal numbers
3. all oops concepts

## **Starkenn Technologies**

1) Round 1&2 - Aptitude test + Domain Test

2) Round - 3 Coding round

- Q. Write code to Find occurrence of no from array
- Q. Write code to Find Max and Min no from array
- Q. Reverse linked list

3) Interview-1 Technical (Round4)

- Q. Introduction
- Q. Explain your code
- Q. Explain your project.
- Q. What CAN protocol can you draw the CAN standard Message Frame
- Q. Explain Arbitration in CAN
- Q. What is I2C protocol
- Q. What PID explain it properly
- Q. In CAN protocol what is the Purpose of RTR bit and Explain Remote Frame
- Q. Explain CAN communication.

Interview -2 Technical (Round 5)

- Q. Introduction
- Q. Explain your engineering project
- Q. How Can you test your project.
- Q. Write Test Cases for your project Explain that properly.
- Q. Explain I2C bus and open collector configuration properly
- Q. Explain CAN protocol Why CAN
- Q. What is your Role in Project

## **Technical Background Discussions**

### **HR Manager and COO (Round -6)**

Introduction

General Technical Discussion regarding Technical Abilities and Working Compatibility.

## **G and D interview.**

### **Booting Linux**

#### **Rtos**

#### **Embedded os**

- 1) Diff b/w embedded os normal os and rtos
- 2) Scheduling algorithm
- 3) Practical application of structure and union.
- 4) I2c spi uart can
- 5) Diff b/w i2c and spi

- 6) Diff b/w i2c and can
- 7) I2c waveforms
- 8) Rtos configuration
- 9) What is svc
- 10) What is systick
- 11) Code to write change the bits
- 12) Code to write difference between different storage class
- 13) Stm booting
- 14) Process vs thread
- 15) Stack
- 16) Queue
- 17) Stack vs queue
- 18) Project implementation and its functionality
- 19) Cortex m architecture
- 20) About interrupt
- 21) Basic shell script
- 22) Queue application
- 23) Semaphore and mutex
- 24) Semaphore vs mutex
- 25) What is race condition
- 26) Adc and resolution
- 27) I2c and 3 slave connection
- 28) I2c errors
- 29) How to find base address of 1d array?
- 30) How to find base address of 2d array?
- 31) Dma

1. What is the difference between Global and Static
2. What is typecasting. Explain with example
3. What is pointer and what is pointer-to-pointer
4. What is dangling pointer and how to avoid it
5. Why C is fastest language
6. Explain recursion with a simple code
7. Explain const and volatile
8. Explain void pointer
9. Explain NULL pointer
10. Difference between macros and const
11. Where are macros and const are stored
12. What are the advantages of C and C++
13. What does core dump contain
14. What is segmentation fault in case of C and OS
15. Code snippet given to solve
16. Why C is mid-level language
17. Write a code to combine contents of two text files into one file
18. Explain code to find substring from a given string
19. How would you compare two integers without using "==" operator?



20. What are the four pillars of OOPS, explain each with an example
21. What is static polymorphism
22. What is virtual function
23. If we declare a function as a friend function then what are the functions and data members that are not accessible
24. What is reference in C++? How they differ from reference in Java
25. Write a query in SQL to fetch content of emp table
26. Explain your project
27. Why did you do this project
28. What was the range of ultrasonic sensor
29. What is IPC and what are the IPC mechanisms? Explain
30. Explain semaphore, mutex and conditional variable
31. Explain spinlock and tasklet.
32. If a variable is accessed by two processes/threads in multithreading environment then how to avoid the deadlock

## CDAC PUNE INTERVIEW QUESTIONS

1. CDAC Project in detail
2. What did you achieve through your project
3. How did you test/demonstrate your Project and what was its outcome
4. What problems did you face in your project
5. Why do we use library functions instead of directly using system calls
6. Difference between Conditional Compilation and Conditional Execution

### SIMD vs MIMD

7. Can a process in Linux run in Windows.
8. Difference between executable file of Windows and Linux? What is magic number
9. Can we change the extension of executable file in Linux? Will it still work the same
10. What is a dangling pointer
11. Difference between void and dangling pointer
12. What is memory leakage
13. Why any pointer is of size 8-bytes
14. How to go from user space to kernel space
15. Difference between fopen() and open()
16. Which is the fastest data structure from which retrieving data is faster
17. In a 64-bit register how would you set the 11th bit? What is the logic behind setting a bit
18. In a 32-bit Data Bus and 64-bit Address Bus, what are the maximum number of addresses possible
19. Difference between I2C and UART and SPI
20. Error Conditions of Communication protocols

21. What is the bit rate of CAN
22. If CAN is used for communication between different subsystems of a vehicle, then is it suitable for communication between media player and different displays available for each passenger.
23. What are the Local Area Network (LAN) protocols
24. What is a re-entrant code
25. What the conditions to be satisfied for Deadlock
26. What if two threads try to access/modify same variable? What happens in a race condition? What is the problem
27. Difference between resistance and impedance
28. What are the capacitor characteristics

**Situation Based: -**

1. If a program is already burned in a microcontroller connected to a sensor and after powering the controller, it powers on but doesn't work. How will you debug the problem? How would you
2. How would you determine the resistance of a resistor connected in a circuit? Will the internal resistance of the battery affect the overall resistance.
3. In a circuit with lots of components how would you find a short in such a circuit
4. Modes of a multi-meter and how can you test a transistor using multi-meter
5. What is a pull-up resistor and why is it used? Does it keep the line always pulled to high voltage
6. What is the driver line configuration for pull-up resistor
7. What is a pull-down resistor and why is it used? What is its internal configuration
8. What are the types of MOSFET configurations
9. How are the terminals of BJT analogous to that of MOSFET

**CDAC Bangalore Interview Questions**

1. Introduce yourself in only technical terms
2. What is semaphore and mutex? What is the difference between them
3. What is a race condition? Explain
4. Explain what is a critical section
5. What is pcb?
6. Explain write function call. Where the data is written when we use this function? -> RAM or HDD??
7. What is a Pipeline? Explain in detail with its hazards
8. What do you mean by Static linking and dynamic linking
9. Difference between singly and doubly linked list
10. If there is one head and it is also the tail then which data structure it is?
11. Have you done coding in device drivers? For which device do you write code?
12. Can you write code of linked list? I said yes. Then asked me only logic.
13. Write a code of string reversal in C
14. What are the Oops concepts
15. What are the storage classes in C and what is its significance.
16. What are the IPC mechanisms in OS
17. Where stack data structure is used

18. Write code for taking 2 different string and then concatenate this 2 strings into 3rd string. You don't know the length.
19. Write a structure for the linked list.
20. Tell me about the CAN protocol
21. `char str[]={'a','b','c'}`, `char str[]="hello"` are this both same or different. if same/different then give the reasons
22. What is RISC and CISC
23. How does a cpu know that a particular number is integer or float. Because in the binary their representation will be
24. Difference between SPI and UART protocol
25. Compilation Flow
26. What is Stack
27. STM32 boards Architecture
28. What is cross compilation

## CDAC Bangalore Interview Questions

1. Tell me about your project
2. What do you mean by static keyword
3. What is the size of a integer data type -architecture and compiler dependent
4. If you have a structure with char, float and int data types what will be the size of the structure when we check its s
5. Why will the structure size be 12 bytes or 9 bytes specifically
6. What is structure padding and what is `#pragma pack()`
7. If you have a structure with char, float and int data types what will be the size of the structure when we check its s
8. What is the difference between static and extern keywords
9. How extern is used
10. What is the scope of static and extern
11. Do you know about BBB.
12. BBB architecture-i didn't know about the architecture so I said I have done cross compilation on BBB.
13. If we can write our code directly on our BBB then why do we write the source code in other machine and then co
14. What do you mean by paging
15. Do you know about MMU
16. What is the working of a MMU unit
17. What is virtual memory
18. What are the conditions where pages are swapped out of memory
19. Are pages swapped out only when memory is full
20. What are different synchronization mechanism
21. What is a critical section?
22. What do we use when there are multiple processes for synchronization and explain how do we do it.
23. What is little endian and big endian
24. What is deadlock and what are the two conditions to make sure that deadlock doesn't happen.
25. When will there be head and tail at the same node in a linked list

26. How can you merge two linked lists together
27. Factors affected when size of any data byte changes (32 bit machine = 2byte)
28. What is meant by context switching
29. Write factorial program in c. and if our output is 157 digit number then what should modify in the code.
30. Write a pattern printing program of (A, BB, CCC, DDDD, EEEEE).
31. Explain process life cycle

## **TOSHIBA SOFTWARE INTERVIEW QUESTIONS**

1. Write a program to print "Hello World" 10 times without using loops.
2. Do void functions return a value?
3. What is inline function?
4. What is difference between inline functions and macro, why are macros preferred over inline?
5. Why use macro if inline functions and constant variables exist?
6. Difference between malloc and new?
7. How to free memory allocated by malloc or new? how it works internally?
8. Write code and algorithm to reverse an array.
9. Write code and algorithm to reverse an array.
10. Algorithm to reverse a linked list.
11. How to find loop in a linear linked list?
12. Code to find if nth bit of a number is set or not.
13. How to check if given number is a power of 2 without using loop.
14. Code to find if a given number is log of 2.
15. What is segmentation and paging, which is better and why?
16. What is cache memory? what is cache hit and cache miss?
17. What is TLB?
18. What is fork?
19. If I run a program with fork, how many processes are running on my system?
20. write down a code for stack implementation (initialization, push, pop)
21. write down a code for linked list initialization and also write down code for delete at position(not given position on
22. which project you have done in PG
23. Explain I2C protocol(because it is in my project)
24. what is binary search tree(code to search left most element of tree)
25. Memory layout of C code
26. what is the difference between thread and process
27. how stack and heap grows
28. How interrupt handler Works
29. what is pseudo character device driver and what it does explain the code of that
30. what is Init process what is pid of that process and what it does ,what is virtual memory

## **Aeron Interview Questions**

1. Toggle the bit?
2. what is core dump?
3. SPI pins
4. multifunction of 2 number without arithmetic operations
5. PWM
6. Voltage divider
7. RS232
8. compilation flow
9. Set and Clear the nth pin
10. Is it required to write return in ISR?
11. Multithreading Concept. And then how synchronization occurs between thread?
12. Which is faster I2C or SPI?
13. Frequency of Bluetooth
14. Difference between Ram and EPROM
15. synchronization mechanism

## **Phi Audiocom System-Interview Questions**

1. Brief introduction about yourself
2. Explain your project and your role
3. explain I2C in detail
4. difference between GPOS and RTOS
5. Explain ISR
6. what is context switching
7. Explain priority inversion and how to resolve it
8. What is RAID?
9. write a c code, in which take string from user and display the first non-repetitive character
10. explain logic behind how you can use mutex in two threads to avoid irrespctive result.